Deep Learning (Assignment Project)

Binary Image Classification using CNN

Project Dataset: Cats and Dogs Classification Dataset

Problem Statement

Image classification is one of the most widely studied problems in computer vision. One common benchmark problem is distinguishing between images of cats and dogs, which requires the model to learn complex visual patterns such as shapes, textures, and edges.

The goal of this project is to develop a **Convolutional Neural Network (CNN)** to perform binary image classification using the **Dogs vs Cats dataset**. The dataset contains thousands of labeled images of cats and dogs with varying backgrounds, lighting conditions, and poses.

The CNN model will automatically learn relevant visual features from raw images and classify each image into one of two categories: **cat or dog**.

The project includes dataset preprocessing, CNN model design, training and validation, and evaluation using metrics such as **accuracy, precision, recall, and confusion matrix**

Expected Outcome

A trained CNN model capable of accurately distinguishing between cats and dogs, typically achieving 80–95% accuracy depending on architecture and preprocessing techniques.

athvikaalm2018@gmail.com